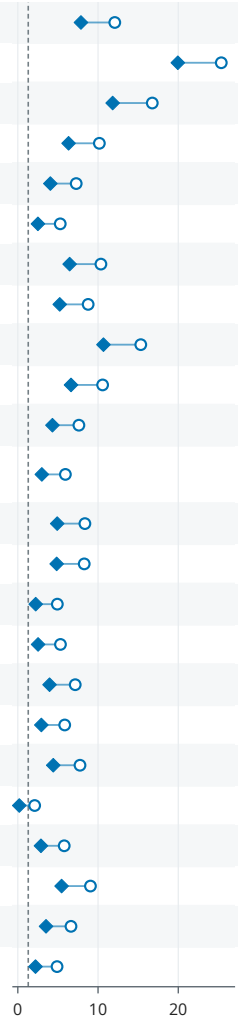
Annotation P value ◆ Annotation FDR ----- P or FDR = 0.05

a Lower expression · 24 modules

Module / top GO:BP or Reactome term	Genes	P value	BH FDR
D01 Interferon-response genes Reactome: interferon signalling	11	8.19e-13	1.35e-08
D02 Integrin–collagen–matrix module Reactome: integrin cell surface interactions	15	4.37e-26	1.14e-20
D03 Ankyrin–spectrin scaffold Reactome: interaction between L1 and ankyrins	6	1.74e-17	1.51e-12
D04 Chemokines, growth factors and contractile genes GO:BP: positive regulation of MAPK cascade	17	6.86e-11	4.70e-07
D05 Contractile / structural genes GO:BP: heart development	5	5.27e-08	8.80e-05
D06 Amino-acid / aldehyde metabolism GO:BP: regulation of catecholamine metabolic process	4	5.14e-06	0.00312
D07 Adhesion / sensory scaffold GO:BP: photoreceptor cell maintenance	4	4.39e-11	3.47e-07
D08 AKR1C enzymes and DHDH GO:BP: progesterone metabolic process	4	1.71e-09	5.94e-06
D09 Immune / matrix signalling GO:BP: positive regulation of immune system process	28	4.71e-16	2.12e-11
D10 Cadherin–junction module GO:BP: cell cell junction assembly	5	2.69e-11	2.34e-07
D11 PI3K–PLC signalling Reactome: signalling by GPCR	5	2.43e-08	4.84e-05
D12 Secreted carrier / plasma proteins Reactome: regulation of insulin like growth factor IGF transport and uptake by insulin like growth factor binding proteins IGFBPs	4	1.18e-06	0.00101
D13 Cyclic-nucleotide metabolism Reactome: nitric oxide stimulates guanylate cyclase	4	4.35e-09	1.23e-05
D14 Thyroid-associated source genes GO:BP: thyroid hormone generation	4	5.28e-09	1.43e-05
D15 UNC5–FLRT–latrophilin module GO:BP: neuron projection guidance	4	1.19e-05	0.0059
D16 BCL11A / BCL11B / TAL1 Reactome: formation of the embryonic stem cell BAF esBAF complex	3	4.83e-06	0.00297
D17 Calcitonin-like receptor module Reactome: class b 2 secretin family receptors	3	6.99e-08	1.10e-04
D18 Innate-response signalling GO:BP: cellular response to molecule of bacterial origin	3	1.43e-06	0.00117
D19 Nicotinic receptor subunits GO:BP: response to acetylcholine	3	1.78e-08	3.82e-05
D20 CREB-family / coactivator genes GO:BP: negative regulation of programmed cell death	3	0.00794	0.657
D21 Ciliary motor genes GO:BP: cilium or flagellum dependent cell motility	3	1.65e-06	0.0013
D22 KLRC receptor-family genes GO:BP: response to lectin	3	8.84e-10	3.49e-06
D23 Sphingolipid metabolism GO:BP: ceramide metabolic process	3	2.36e-07	3.04e-04
D24 Trafficking / actin-associated genes Reactome: retrograde neurotrophin signalling	6	1.28e-05	0.00625

b Higher expression · 18 modules

Module / top GO:BP or Reactome term	Genes	P value	BH FDR
U01 Nicotinic receptor subunits Reactome: acetylcholine binding and downstream events	5	8.47e-17	5.51e-12
U02 Mucin / CDX2 module Reactome: dectin 2 family	5	2.33e-09	7.23e-06
U03 Chemokine receptors and immune co-signalling Reactome: chemokine receptors bind chemokines	7	9.03e-11	5.74e-07
U04 Calcium-channel module GO:BP: calcium ion import across plasma membrane	4	2.54e-08	4.94e-05
U05 Hedgehog-associated module Reactome: hedgehog on state	4	5.19e-10	2.36e-06
U06 Eph–ephrin module GO:BP: ephrin receptor signalling pathway	4	1.04e-10	6.26e-07
U07 Eicosanoid metabolism, glutathione enzymes and cytokines GO:BP: arachidonate metabolic process	11	1.28e-08	2.98e-05
U08 WNT ligands / modulators GO:BP: canonical WNT signalling pathway	4	1.34e-07	1.86e-04
U09 Synaptic adhesion-associated genes Reactome: neurexins and neuroligins	4	1.13e-10	6.55e-07
U10 Potassium-channel module Reactome: inwardly rectifying K channels	3	5.68e-09	1.53e-05
U11 BMP ligands / antagonist GO:BP: response to BMP	3	8.14e-07	7.50e-04
U12 Pigmentation-associated genes Reactome: regulation of MITF m dependent genes involved in pigmentation	3	2.07e-08	4.21e-05
U13 Actin-associated genes GO:BP: actin filament based process	3	1.24e-04	0.0346
U14 piRNA-associated genes GO:BP: piRNA processing	3	5.56e-10	2.44e-06
U15 p75-related receptor module Reactome: p75NTR regulates axonogenesis	3	1.27e-06	0.00107
U16 TREM receptors and CKLF/CMTM family GO:BP: taxis	6	3.41e-08	6.26e-05
U17 A distinct matrix-associated module Reactome: degradation of the extracellular matrix	10	4.88e-16	2.12e-11
U18 Netrin–guidance module GO:BP: neuron projection guidance	4	4.28e-08	7.44e-05



Annotation BH adjustment: 42 modules × 6,203 terms; expressed STRING-mapped background.